The Glove Problem

Spring 2022 ARML Power Contest

The Solutions

1. Each doctor or patient must touch two glove surfaces that have never previously touched any other person. Since a pair of gloves has four surfaces (two on each glove), each person must use "half" a pair of gloves that no one else uses. Because pairs of gloves cannot be used apart from each other, if there is a pair of gloves for which only one surface of each glove was used, the whole pair is considered used, so the number of pairs must be rounded up to the next whole number. Thus $\lceil \frac{m+n}{2} \rceil$ pairs of gloves is the lower bound.

2. Since person A_{i-1} would touch A_i with glove G_{i-1} on the right hand, the outside of glove G_{i-1} is safe for A_i to touch, and thus A_i could wear glove $\overline{G_{i-1}}$ on their left hand to touch person A_{i-1}. Then glove $\overline{A_{i-2}}$ could be placed over it if A_i needs to touch A_{i-2}, and so on. So layer $\overline{A_{i-1}\, A_{i-2} \cdots A_j}$, the sequence being treated modulo t if $j > i$.

3. There are several possibilities. One is to use the three-doctor-one-patient protocol, using just two pairs of gloves, giving the third pair of gloves to the second patient to protect themselves with at all times. For example:

    1. $D_1\, G_1\, G_2\, P_1$
    2. $D_1\, G_1\, G_3\, P_2$
    3. $D_2\, G_2\, P_1$
    4. $D_2\, G_3\, P_2$
    5. $D_3\, \overline{G_1}\, G_2\, P_1$
    6. $D_3\, \overline{G_1}\, G_3\, P_2$

4. For every doctor-patient interaction in a given protocol, simply list the entire interaction backward, turning each glove inside-out (and each pair that was inside-out back to being right-side-in). Then it would appear the patients are operating on the doctors, and every interaction that was safe before continues to be safe. So the roles of doctors and patients can be reversed using the same number of pairs of gloves.

5. (a) **Step 1:** Each doctor must operate on x patients and each patient operated upon by each of $2y$ doctors. So just from one patient's point of view, the makespan must be at least $2y$, and from a doctor's point of view the makespan must be at least x. The makespan will thus be at least $\boxed{\max(x, 2y)}$. But this is easily achieved. For assuming $x \geq 2y$ then just assign the doctors to the first $2y$ patients, and each hour every doctor moves on to the next higher-numbered patient, modulo x. If $x < 2y$ then have each patient visit one of the first x doctors, and each hour every patient moves to the next higher-numbered doctor, modulo $2y$.

Step 2: The single pair J of gloves is used in *every* operation in this step, and there are xy operations to perform. So the makespan of this step is $\boxed{xy}$.

Step 3: The description here is exactly the same as in the first step, except that there are only x patients and y doctors, so this step takes $\boxed{\max(x, y)}$ hours.

Step 4: Again, J must be used in every one of the $\boxed{xy}$ operations.

Step 5: Identical to step 3, so a final $\boxed{\max(x, y)}$ hours.

(b) For a reason similar to that given in **Step 1** above, the makespan must be at least $\boxed{\max(2x, 3y)}$ hours because that is how long it will take any single doctor (if $3y \geq 2x$) or patient (if $2x > 3y$) to complete all operations. And for a similar reason this lower bound is achievable by simply lining up all the patients and doctors and each hour having each doctor move on to the next patient.

6. During **Step 1**, treat the extra doctor as if belonging to group A, wearing the J gloves. Notice that the outside of the J gloves are still safe to use after this step, and the extra doctor has operated on all the P patients. The extra doctor now temporarily gives up the J gloves, which are used to protect the insides of the G gloves in **Step 2**. The inside of J was already contaminated, as were the (now) outsides of the $\overline{I}$ gloves, so no extra contaminations occur at this stage. Perform **Step 3** and **Step 4** as in the protocol. Note that the (now) insides of $\overline{J}$ are still safe, so let the extra doctor wear them again (now inside out from how they were worn in **Step 1**) and treat the extra doctor as if belonging to group B to operate on all the Q patients.

7. Simply give the new pair of gloves to the second extra doctor. This doctor will operate on all the P patients before the protocol outlined above begins, and then finish by operating on all the Q patients after the above protocol finishes.

8. Simply give the new pair of gloves to the extra patient and treat the patient as if they were in group P. The only difference being that when the rest of the P patients give their gloves to the Q patients in **Step 3** this extra patient need not give up the pair of gloves—they are not needed. Alternately, think of adding 1 to x wherein the protocol uses only one extra pair of gloves to allow *two* more patients, and just ignoring the second extra patient!

9. There are six possibilities, depending on the remainder of m on division by 3 and on n by 2. The cases are:

   - $m = 3y$ and $n = 2x$. Then $\lceil \frac{2}{3}m + \frac{1}{2}n \rceil = 2y + x$ and it is given that there is a protocol using this many gloves in the paragraph preceding this problem.
   - $m = 3y + 1$ and $n = 2x$. Then $\lceil \frac{2}{3}m + \frac{1}{2}n \rceil = 2y + 1 + x$ and the protocol found in problem 6 uses this many gloves.
   - $m = 3y + 2$ and $n = 2x$. Then $\lceil \frac{2}{3}m + \frac{1}{2}n \rceil = 2y + 2 + x$ and the necessary protocol was found in problem 7.
   - $m = 3y$, $n = 2x + 1$. Then $\lceil \frac{2}{3}m + \frac{1}{2}n \rceil = 2y + x + 1$, and the existence of the necessary protocol is given in the paragraph preceding this problem.

- $m = 3y+1$, $n = 2x+1$. Then $\lceil \frac{2}{3}m + \frac{1}{2}n \rceil = 2y+x+2$, and the necessary protocol is furnished by the solution to problem 8.
- $m = 3y+2$, $n = 2x+1$. Then $\lceil \frac{2}{3}m + \frac{1}{2}n \rceil = 2y+x+2$, and the existence of the necessary protocol is given in the paragraph preceding the problem.

10. Given the symmetry between patients and doctors as proven in problem 4, this result follows immediately.

11. For four patients and six doctors, $x = y = 2$. In **Step 1** patients P_1 and P_2 contaminate the outside of gloves G_1 and G_1, so arrows start at these patients corresponding to these gloves. Likewise, there are arrows emanating from the two A-doctors with H-gloves, and B-doctors with I-gloves.

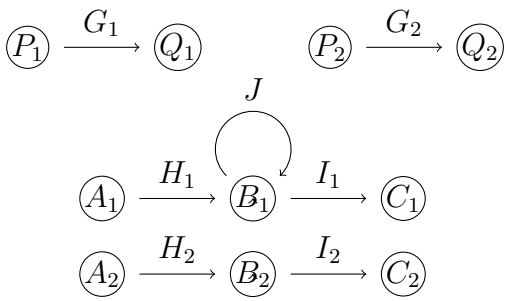
    In **Step 2** the I-gloves are turned inside-out and given the the C-doctors. This contaminates the other side of the I-gloves, so the arrow representing them terminate at the C-doctors. The very first operation (which could be either of the C-doctors operating on either of the P-patients) in this step will contaminate the inside of glove J. Since this happens with the outside of an I-glove, the inside of glove J is actually first contaminated by a B-doctor, say B_1.

    Now **Step 3** has the G-gloves being turned inside-out and given to the Q-patients who are now the first to contaminate their other side. So the arrows representing the G-gloves terminate at the Q-patients. After this, all contacts occur between gloves that are already contaminated, so no arrows will start or end at this stage.

    For **Step 4** no new contaminations occur. At **Step 5** the H-gloves become contaminated on their other side by the B-doctors, so those arrows will end at the B-doctors.

    Finally, since one side of pair J is never contaminated, by the instructions for creating diagrams the arrow representing J returns to the person it started from.

    There are, of course, many ways to draw the diagram just described. One way is shown below:



12. The first time that a doctor operates, or the first time that a patient is operated upon, that person must touch a side of a pair of gloves that has never been touched by another patient or doctor, or by an already-contaminated side of a glove. If this is the first time that pair of gloves has been used, then the arrow representing those gloves will originate at that person; if the other side of the gloves has previously been used then the arrow will terminate at that person; if both sides of the gloves are being used for

the first time in this operation then the arrow will either begin or end with this person, the choice being made at random. In any event, every doctor and patient must have at least one arrow originating or ending with them.

13. Let the arrow from s to t represent pair G of gloves. Since there is no other arrow going into or coming out of either s or t, every operation involving s or t must use this pair of gloves. And the first time that s uses these gloves can be no later than the first time that t uses these gloves because the arrow starts at s. Assume there are people x and y (who might be the same person) so that an operation involving t and y occurs before one involving s and x. Since every operation involving t uses G and the contamination of the s side of G occurs at the same time or before the time that the t side of G becomes comtaminated, the s side of G was already contaminated at the time of the operation between t and y. That means that y had to be protected by some glove(s) other than G. Consider the glove touching the s side of G during the operation between t and y.

    - If that glove was already contaminated, it can't be by s because no other arrow goes into or out of s. But then it will contaminate the s side of G with some other contaminant, and s will not be able to use this glove in the later interaction with x.
    - If that glove was not already contaminated, it will become contaminated by s upon touching the s side of G during the operation between y and t. That means another arrow must either begin or end at s.

    Since all possibilities fail to allow the necessary operation to take place, the ty-operation cannot occur before the sx-operation.

14. (a) Let $D_1 \to D_2$ and $P_1 \to P_2$ be 2-components connecting two doctors and two patients respectively. From problem 13, since $D_1 \to D_2$ is a 2-component, D_1 must operate on P_2 before D_2 can operate on anyone at all. But again by problem 13, since $P_1 \to P_2$ is a 2-component, P_1 must be operated on by D_2 before P_2 can be operated on by anyone. This is a contradiction, so these two 2-components cannot exist within the same diagram.

    (b) If there are three or more 2-components connecting a doctor and a patient, by the pigeonhole principle there are at least two where the arrow "goes the same way." That is, there are at least two components where the arrow goes from doctor to patient, or two where the arrow goes from patient to doctor. Without loss of generality assume the former, and there are 2-components $D_1 \to P_1$ and $D_2 \to P_2$. Then, using problem 13 and the $D_1 \to P_1$ component, the operation between D_1 and P_2 must occur before the operation between D_2 and P_1. But from the $D_2 \to P_2$ component, the operation between D_2 and P_1 must occur between that between D_1 and P_2, again a contradiction. So there are at most two 2-components connecting a patient to a doctor (and these two must have their arrow going in opposite directions).

15. The number of pairs of gloves needed for a protocol is the number of arrow in its diagram. Since m and n are fixed in the problem, the goal is to determine a lower bound based on m and n for the number of arrows. Using the information in the paragraph preceding the problem the ratio of arrows to people in a component is at least $2/3$ unless the component is a 2-component. So to minimize the number of arrows in the diagram start by maximizing the number of people who are part of 2-components.

    By problem 14(a) there cannot be a 2-component consisting of doctors and one consisting of patients. So to maximize the number of 2-components, since there are at least as many patients as doctors, put all the patients except for two into 2-components. Those two can be put into 2-components with doctors, as in problem 14(b). So the maximum number of people in 2-components is $n+2$ if n is even, and $n+1$ if n is odd.

    The best that can be done for the remaining doctors is to form groups of three, each of which will use two pairs of gloves.

    In this way, the minimum number of arrows that can be used is $\frac{1}{2}(n+2)+\frac{2}{3}(m-2) = \frac{1}{2}n+1+\frac{2}{3}m-\frac{4}{3} = \frac{1}{2}n+\frac{2}{3}m-\frac{1}{3}$. If this number is not an integer because n is not even or because $m-2$ is not divisible by 3 then any extra "fractions" of gloves must be rounded up to the next whole number. This gives the desired result.

16. The two sides of the equation are equal unless the the fractional part of $\frac{1}{2}n+\frac{2}{3}m$ is strictly larger than zero and is at most $\frac{1}{3}$. This, in turn, happens only the n is even and m has remainder of 2 on division by 3, or when m is odd and n has remainder 1 on division by 3.

Here is the protocol alluded to several times in the problem allowing $2x$ patients and $3y$ doctors to use just $x+2y$ pairs of gloves. The trick is to strategically use other pairs of gloves when J is needed in the protocol given in the introductory notes. The protocol will also allow all the doctors to operate on each other!

As before, denote the doctors by A_i, B_i, and C_i, as i ranges from 1 to y, and the patients by P_j and Q_j as j ranges from 1 to x. Denote the $x+2y$ pairs of gloves by G_j, H_i, and I_i as i ranges from 1 to y and j from 1 to x. Furthermore, if the protocol lists operations between doctors D_1 and D_2 assume both operations where D_1 operates on D_2 and then D_2 wearing the same gloves in reverse order operates on D_1 are performed.

Step 1: Give the G-gloves to the P-patients, and the H- and I-gloves to the A- and B-doctors. Perform all operations of the form $A_i\,H_i\,G_j\,P_j$, $B_i\,I_i\,G_j\,P_j$, $A_{i_1}\,H_{i_1}\,\overline{H_{i_2}}\,A_{i_2}$, $B_{i_1}\,I_{i_1}\,\overline{I_{i_2}}\,B_{i_2}$, $A_{i_1}\,H_{i_1}\,\overline{I_{i_2}}\,B_{i_2}$. The last three types of operations have allowed all the A- and B- doctors to operate on each other, while the first two allow all A- and B-doctors to operate on all P-patients.

Step 2: Doctors B_2 through B_y give their I-gloves (inside-out, of course) to doctors C_2 through C_y. Now perform the operations $B_1\,I_1\,\overline{H_1}\,\overline{I_j}\,C_j$, $A_i\,H_i\,\overline{H_1}\,I_j\,C_j$, and $C_j\,\overline{I_j}\,H_1\,G_k\,P_k$, with i and j running from 2 to y and k from 1 to x. Glove H_1 has been acting as the J glove, protecting the insides of the G gloves and the outsides of gloves H_2 through H_y,

as well as the outside of I_1. At this point, The P-patients have been operated on by all doctors except C_1, and all doctor-doctor operations have been performed between group A and group C that do not involve either A_1 or C_1. Also, B_1 has operated on all other doctors except C_1.

Step 3: The single operation $C_1\, G_1\, P_1$. P_1 has now been operated upon by all doctors.

Step 4: The operations $C_1\, G_1\, H_1\, G_k\, P_k$, k running from 2 to x. Now C_1 has operated on all P-patients, so the P-patients have completed their operations. Gloves G_2 through G_x continue to have been protected on their inside by H_1.

Step 5: Allow C_1 to complete operations on the A doctors: perform all $C_1\, G_1\, H_1\, \overline{G_i}\, A_i$ for i running from 2 to y. The outsides of H_2 through H_y are still uncontaminated, and doctors A_2 through A_y have traded operations with all other doctors.

Step 6: The single operation $B_1\, I_1\, \overline{H_1}\, \overline{G_1}\, C_1$. The outside of H_1 and I_1 remain uncontaminated.

Step 7: Have patients P_2 through P_y give their G-gloves to Q_2 through Q_y. Now do the operation $B_1\, I_1\, \overline{H_1}\, \overline{G_k}\, Q_k$ and $A_i\, H_i\, \overline{H_1}\, \overline{G_k}\, Q_k$, i from 2 to y and k from 2 to x. That is, B_1 and all A-doctors except A_1 operate on all the Q-patients except Q_1.

Step 8: The single operation $B_1\, I_1\, Q_1$. Note that the outside of I_1 has been kept free of contamination up to this point so is safe to use on Q_1. Doctor B_1 has now operated on all patients and all doctors.

Step 9: All the operations $A_i\, H_i\, \overline{H_1}\, I_1\, Q_1$. Doctors A_2 through A_y have now operated on everyone and the outsides of their H-gloves are still uncontaminated. So give them to doctors B_2 through B_y.

Step 10: Even though it has been used in so many operations, H_1 has been used to protect other gloves and remains with its outside uncontaminated. Turn it inside out and give it to doctor A_1. Doctors A_2 through A_y are done. Doctor B_1 is also done, while doctors B_2 through B_y are wearing the inside-out $\overline{H}$-gloves. Doctor C_1 is currently wearing G_1, while doctors C_2 through C_y are wearing the inside-out $\overline{I}$-gloves. All P-patients are finished, Q_1 is using the outside of pair I_1, and the rest of the Q-patients are using the original insides of the G-gloves. Thus, everyone who still needs to operate or to be operated on has their own set of gloves, and all operates can proceed.

Eliminating the remaining cases where the minimum might be less than $\lceil \frac{1}{2}n + \frac{2}{3}m \rceil$ due to the extra $-\frac{1}{3}$ in the formula is a matter of *a lot* of casework and will not be shown here.

The problem has more history than just its connection to Martin Gardner, Isaac Asimov, and Edgar Rice Burroughs. Soon after it was published in the three-doctor-one-patient case, mathematicians and puzzle-lovers immediately tried to generalize, looking for the minimum number of pairs of gloves necessary to solve the problem for general numbers of patients and doctors, and in cases where some of the doctors are also patients and need to operate on each other as well. There are three main references about the topic: a paper by Hajnal and

Lovasz (which is hard to obtain), a paper by Orlitsky and Shepp (which is nearly impossible to obtain) and a chapter from a book by Vardi (which is long out-of-print). Oddly, the first two of these don't mention patients and doctors and gloves. The first discusses sick rabbits who must be exposed to various radioactive samples while being protected by membranes; the second discusses computers in a network that must communicate without exchanging computer viruses, so do so via protection interfaces. Why the various settings?

The third source goes back to the *original* setting of the problem, before Gardner's doctors and patients. Which was a group of men and a group of women, each of whom wanted to have sexual relations with each member of the opposite sex, using a condom as protection from sexually transmitted diseases. But the prudery of the late 1960's and early 1970's prevented the authors from using that setting! In fact, Vardi notes that he was almost prevented from publishing using that setting by a woman who was a junior editor at the publishing company deemed the material too sexist—until another woman who was a senior editor pointed out that the situation was completely symmetric in the men and women so was not sexist at all! In light of this being a contest for students, it was thought most prudent to return to the doctor-patent-glove setting.

Interestingly, the problem concerning the makespan of the protocol was inspired by the *Wikipedia* article on the problem (in its condom form). The material on the makespan being much shorter if just a few more resources are available is correct, but the rest of the article is mostly incorrect, claiming a lower bound of $m+n-2$ instead of $\lceil \frac{1}{2}n + \frac{2}{3}m \rceil$, which is nearly 70% too large! This will be corrected when an article by the problem author is published, thus having a reference to base the correction on!